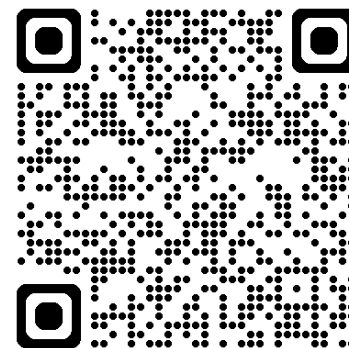




深度學習 Deep Learning

Term Project Introduction

Instructor: 林英嘉 (Ying-Jia Lin)
2025/10/15



[Course GitHub](#)



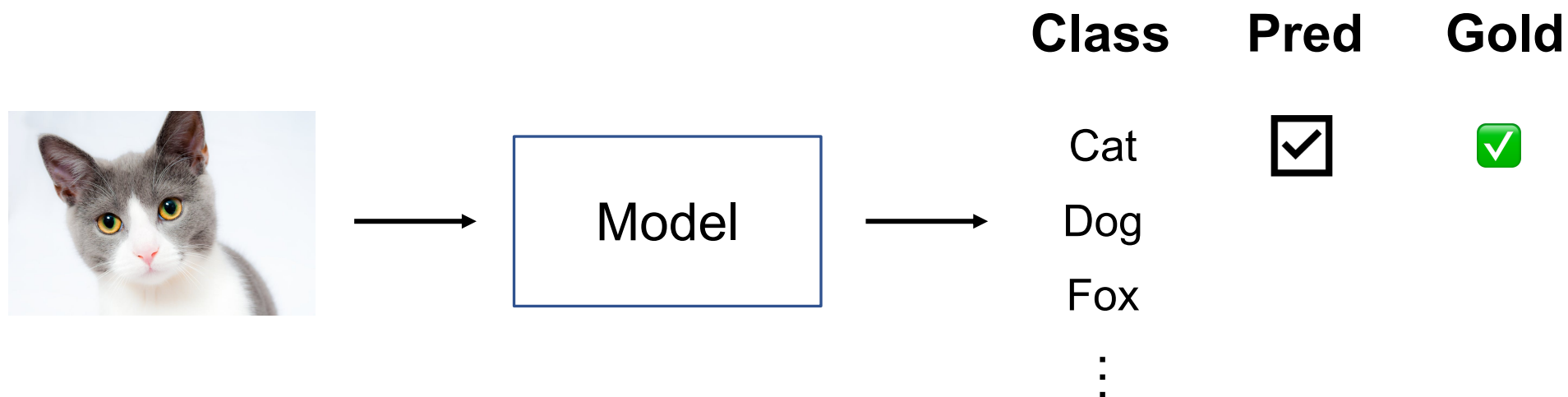
[Slido # DL1015](#)

Final Project 是什麼？

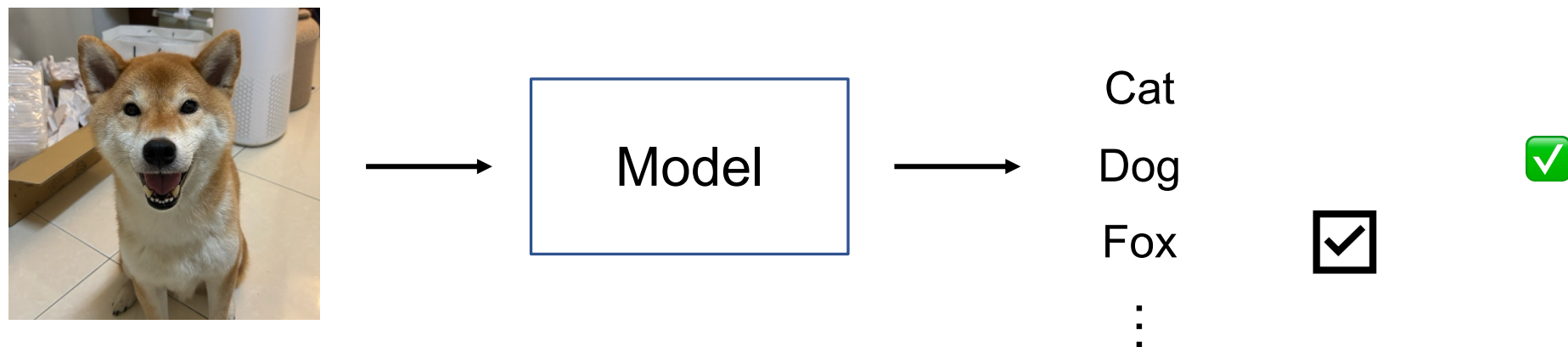
- Final Project 是一種完整的訓練
- 你將自行從頭到尾做完一個題目，包含：
 - 資料前處理 (Data Pre-processing)
 - 模型訓練 (Model Training)
 - 模型評估 (Evaluations) --> 了解問題並改進模型
- 每組人數待修課同學總數訂定，原則上1-3人/組



如何評估機器學習模型的效能？

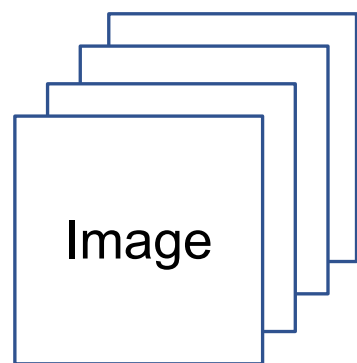


我們不能只憑一筆資料就說模型很棒...



自動化評估

- 「評估」的意義：量化模型的表現



資料



(假設用二分類任務)

	Pred	Gold
1	1	0
2	1	1
3	1	1
4	1	1

正確率：0.75

真正答對的數量 /
總預測的數量



評估方式 (計算模型有多高分的方式)

- 不同任務有不同的評估方式

類型

Automatic Evaluation
(自動化評估)



Human Evaluation
(人工評估)



「生成」任務較為需要



What is Kaggle?

- Kaggle is a platform that provides:
 - Real-world datasets for machine learning
 - Competitions with prizes (sometimes with money)
 - Discussion forum with **a lots of code examples**

kaggle



Kaggle submission types

- Traditional competitions 本課程 Term project 採用此方式
 - Upload **submission file** (e.g., *.csv)
- Code / Notebook competitions
 - Upload **code** (e.g., *.ipynb)



If you play Kaggle a lot ...

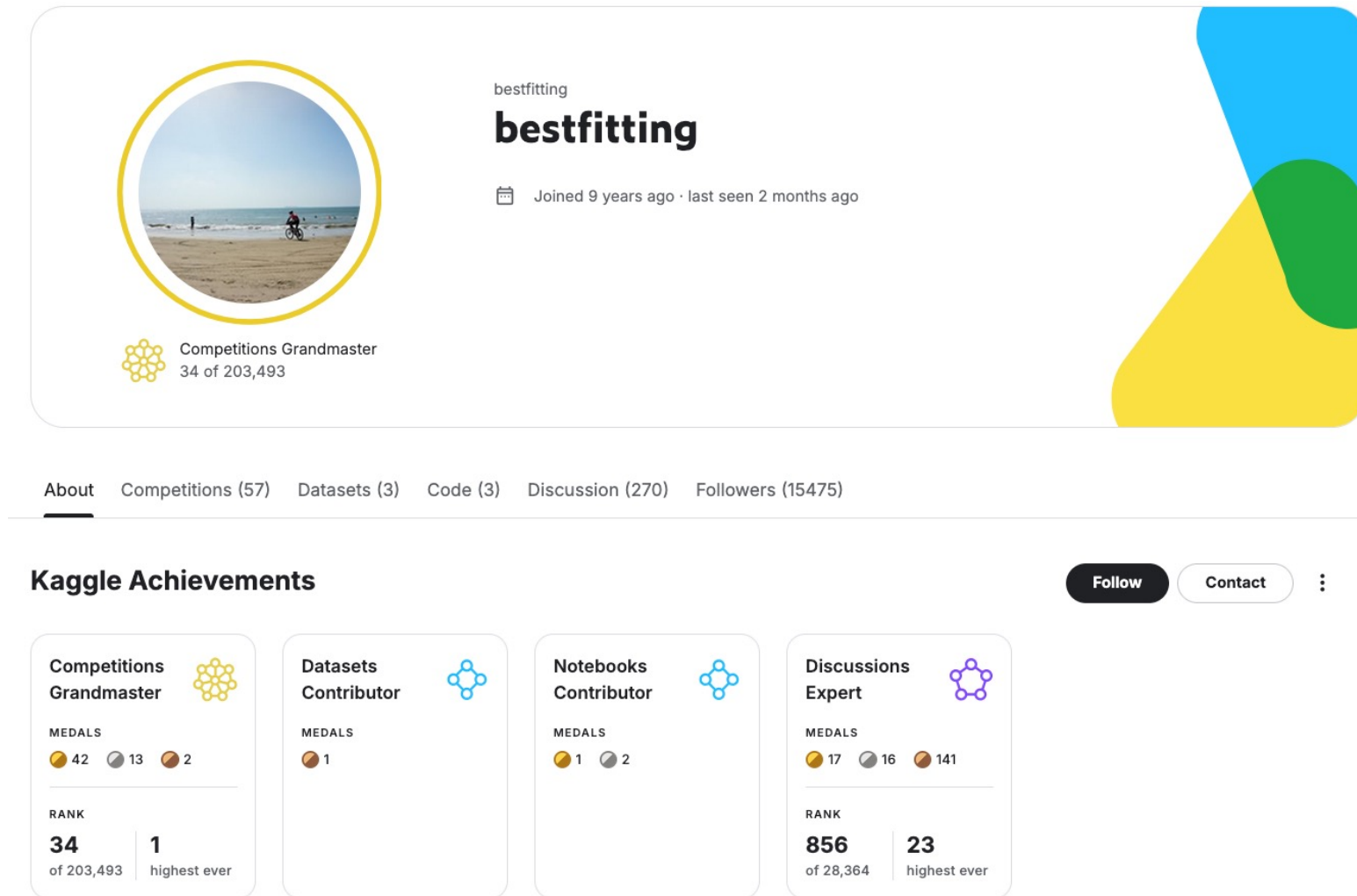


Figure source: <https://www.kaggle.com/bestfitting>

Outline of tasks

Platform	Competition Name	Data Type	Task Type	Why special? (Difficulty)
Kaggle	Dog Breed Identification	Image	Image Classification	120 classes
	Plant Pathology 2020 - FGVC7	Image	Image Classification	Class imbalance
	Natural Language Processing with Disaster Tweets	Text	Text Classification	NLP
	Wikipedia - Image/Caption Matching	Image / Text	Retrieval	CV+NLP



Dog Breed Identification

- <https://www.kaggle.com/competitions/dog-breed-identification/overview>

000bec180eb18c7604dcecc8fe0dba07.jpg (54.78 kB)



Data Explorer

750.43 MB

- test
- train
- ▮ labels.csv
- ▮ sample_submission.csv

Summary

- 20.6k files
- 123 columns

[Download All](#)



Dog Breed Identification

affenpinscher
afghan_hound
african_hunting_dog
airedale
american_staffordshire_terrier
appenzeller
australian_terrier
basenji
basset
beagle
bedlington_terrier
bernese_mountain_dog
black-and-tan_coonhound
blenheim_spaniel
bloodhound
bluetick
border_collie
border_terrier
borzoi
boston_bull
bouvier_des_flandres
boxer
brabancon_griffon
briard
brittany_spaniel

⋮



000bec180eb18c7604dcecc8fe0dba07.jpg (54.78 kB)



labels.csv (482.06 kB)

Detail Compact Column

↻ id	≡	▲ breed	≡
10222 unique values		scottish_deerhound 1% maltese_dog 1% Other (9979) 98%	
000bec180eb18c7604dcecc8fe0dba07		boston_bull	
001513dfcb2ffafc82cccf4d8bbaba97		dingo	
001cdf01b096e06d78e9e5112d419397		pekinese	
00214f311d5d2247d5dfe4fe24b2303d		bluetick	
0021f9ceb3235effd7fcde7f7538ed62		golden_retriever	
002211c81b498ef88e1b40b9abf84e1d		bedlington_terrier	
00290d3e1fdd27226ba27a8ce248ce85		bedlington_terrier	

Dog Breed Identification (evaluation)

- 評估指標為 Multi Class Log Loss，值越小越好
 - 類似 Cross-entropy
 - 預測正確的話 loss 為零

$$- \boxed{1} \times \log(\boxed{0.1}) = 1$$



Plant Pathology 2020 - FGVC7

- <https://www.kaggle.com/competitions/plant-pathology-2020-fgvc7/data?select=images>

images (3642 files) ⌵ >



Test_0.jpg
232.54 kB



Test_1.jpg
301.02 kB



Test_10.jpg
250.57 kB



Test_100.jpg
281.3 kB



Test_1000.jpg
243.87 kB



Test_1001.jpg
230.99 kB

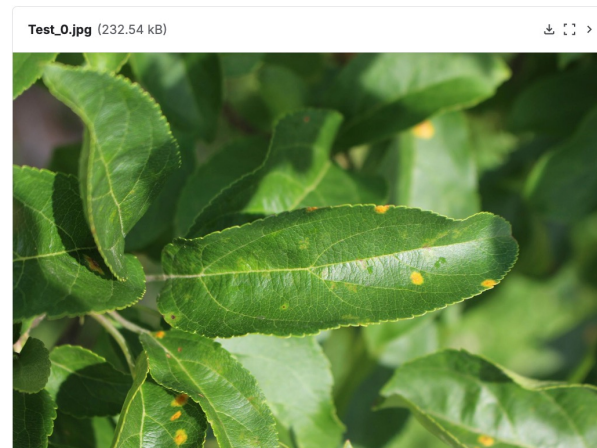
Data Explorer
823.79 MB

- images
 - sample_submission.csv
 - test.csv
 - train.csv



Plant Pathology 2020 - FGVC7

- 每張影像有4個類別
 - Healthy
 - Multiple diseases (多重疾病)
 - Rust (鏽病)
 - Scab (疥瘡病)



train.csv (33.53 kB) 📄 🔍 >

Detail Compact Column 5 of 5 columns ▾

▲ image_id ▾	# healthy ▾	# multiple_diseases ▾	# rust ▾	# scab ▾
1821 unique values	 0 1	 0 1	 0 1	 0 1
Train_0	0	0	0	1



Plant Pathology 2020 - FGVC7 (Evaluation)

Evaluation

Submissions are evaluated on **mean column-wise ROC AUC**. In other words, the score is the average of the individual AUCs of each predicted column.

針對每個 label 的模型預測各算一次 AUC (Area Under the Curve) 之後，取平均



ROC Curve

(Useful!!) <https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc?hl=zh-tw>

ROC curve: Receiver operating characteristic curve (接收者操作特徵曲線)

ROC curve 定義: 針對不同的模型分類門檻下的模型表現圖

- X軸 : False positive rate (FPR); Y軸 : True positive rate (TPR)
- AUC: ROC 的底面下面積 (越大越好)

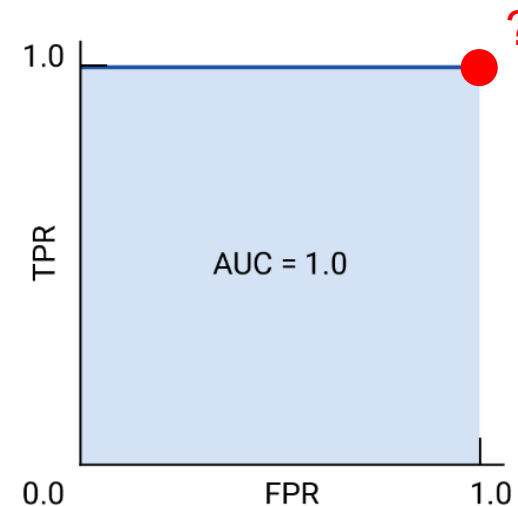
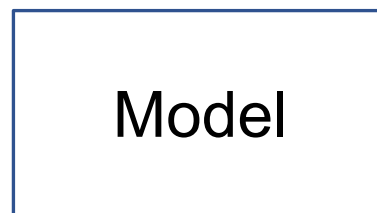
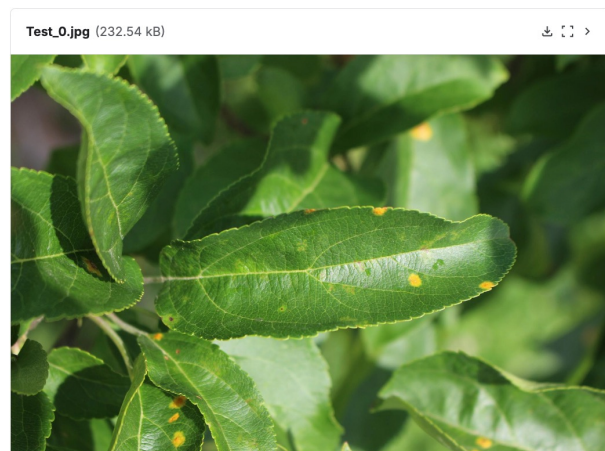


圖1 : ROC 和 AUC , 這是一個完美假設的模型。



模型分類門檻 (Threshold)



機率 = 0.55

Rust

模型輸出

答案

1

我們應該設多少門檻？

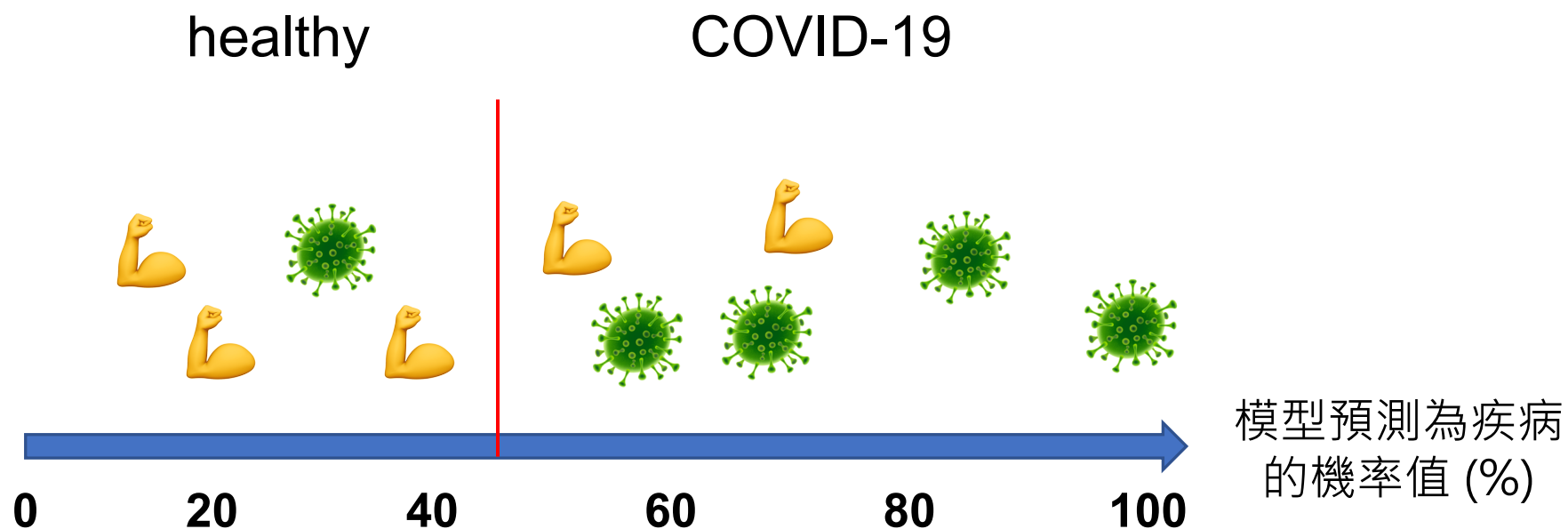
threshold = 0.5 -> 模型預測為1，正確數量+1

threshold = 0.6 -> 模型預測為0，正確數量-1



TPR and FPR 範例

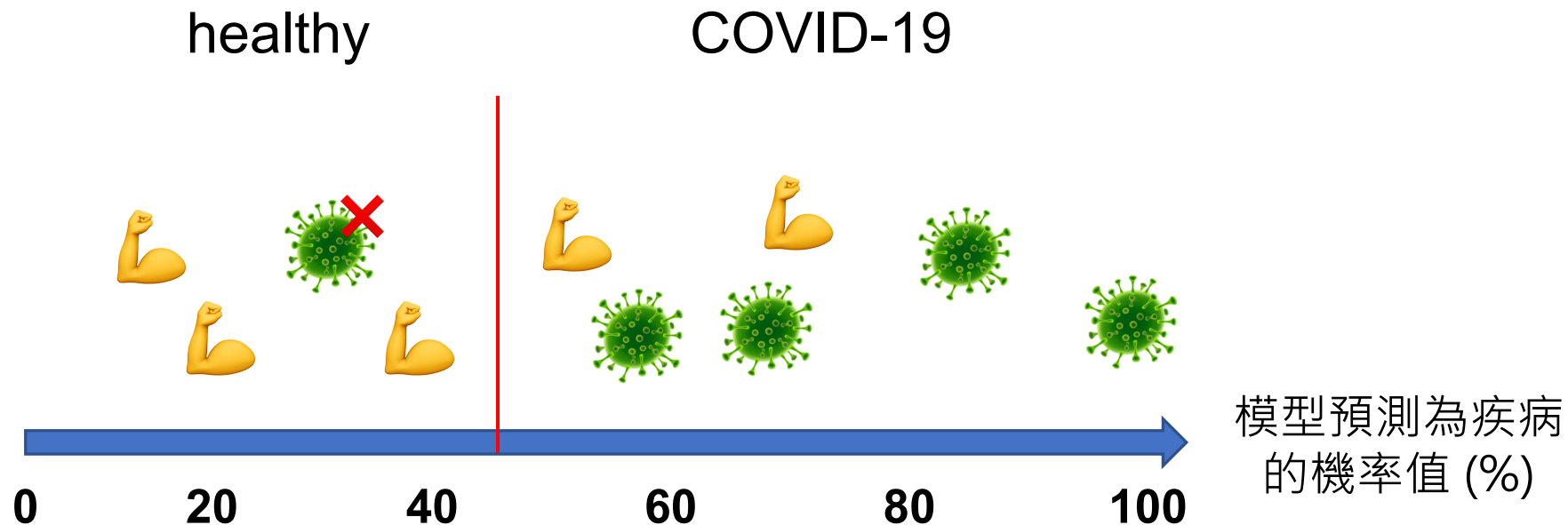
假設我們用模型對10個人做COVID-19檢測：
(COVID-19 代表 positive)



TPR and FPR 範例 (TPR)

假設我們用模型對10個人做COVID-19檢測：
(healthy 代表 negative; COVID-19 代表 positive)

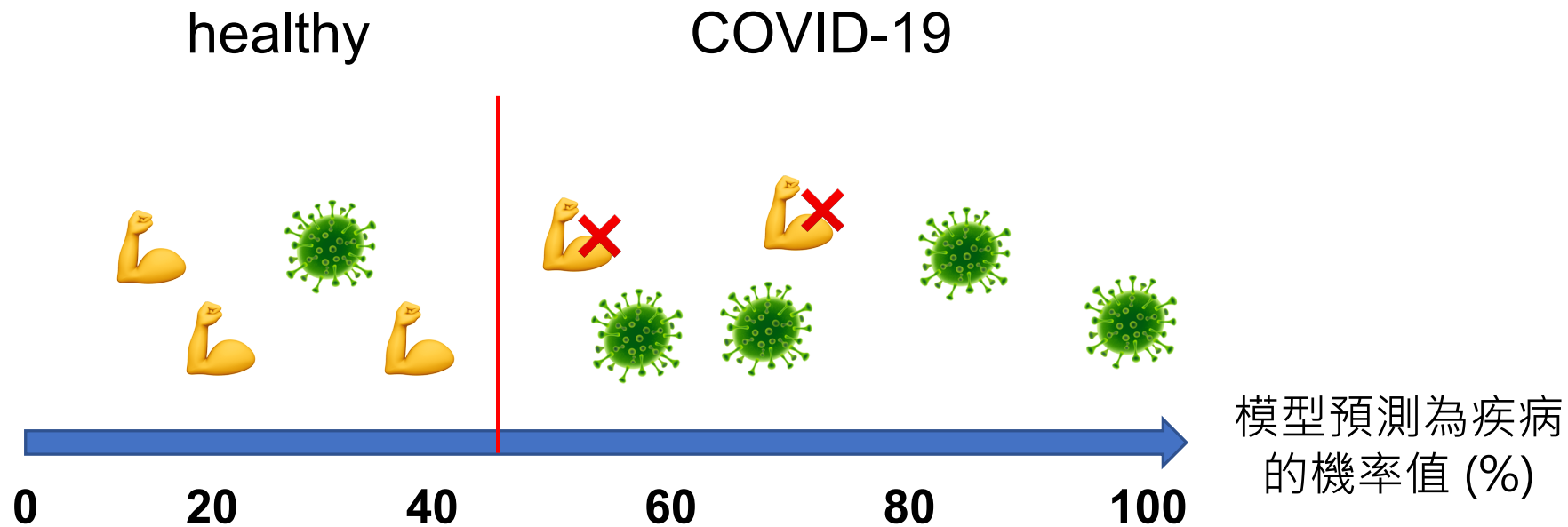
True positive rate
(TPR) = $4 / 5 = 0.8$



TPR and FPR 範例 (FPR)

假設我們用模型對10個人做COVID-19檢測：
(healthy 代表 negative; COVID-19 代表 positive)

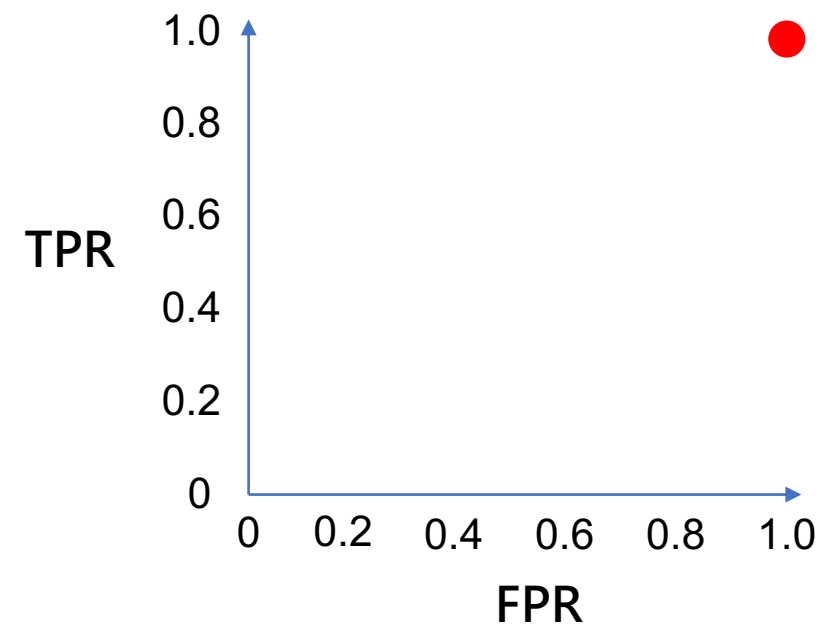
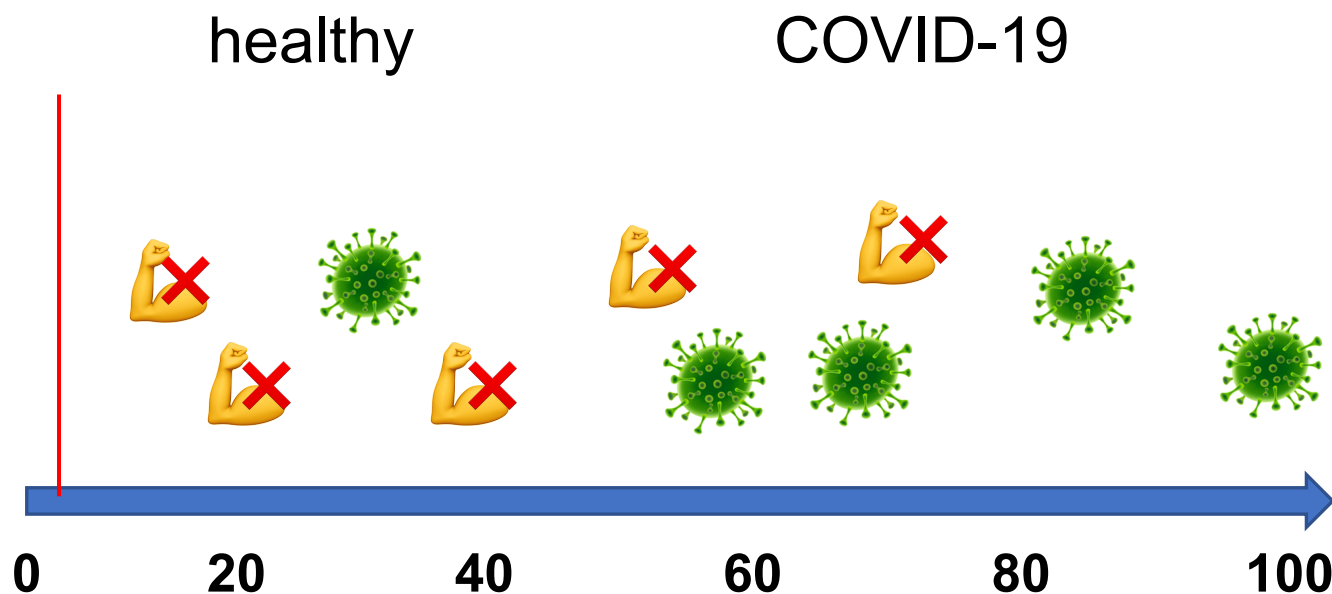
False positive rate
(FPR) = $2 / 5 = 0.4$



ROC Curve

假設我們用模型對10個人做COVID-19檢測：
(healthy 代表 negative; COVID-19 代表 positive)

TPR: 1.0, FPR: 1.0



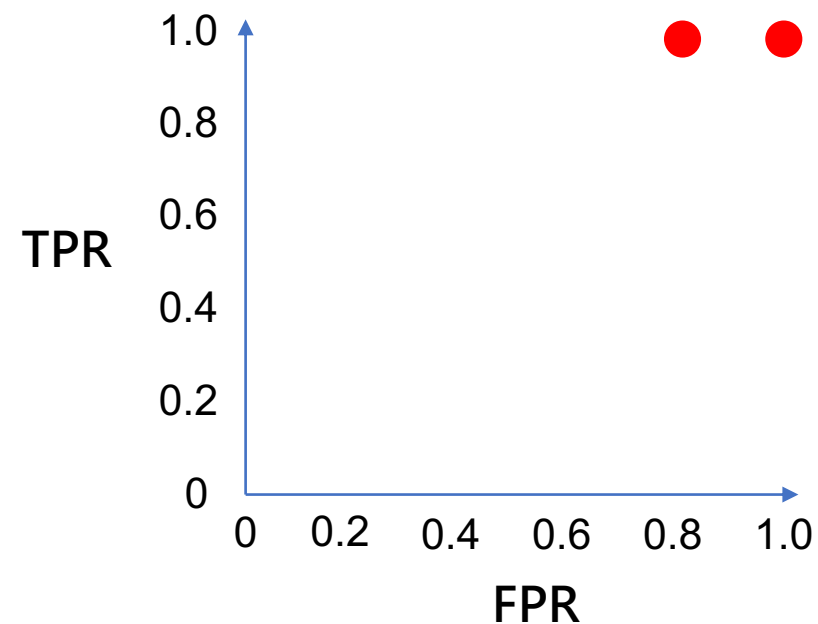
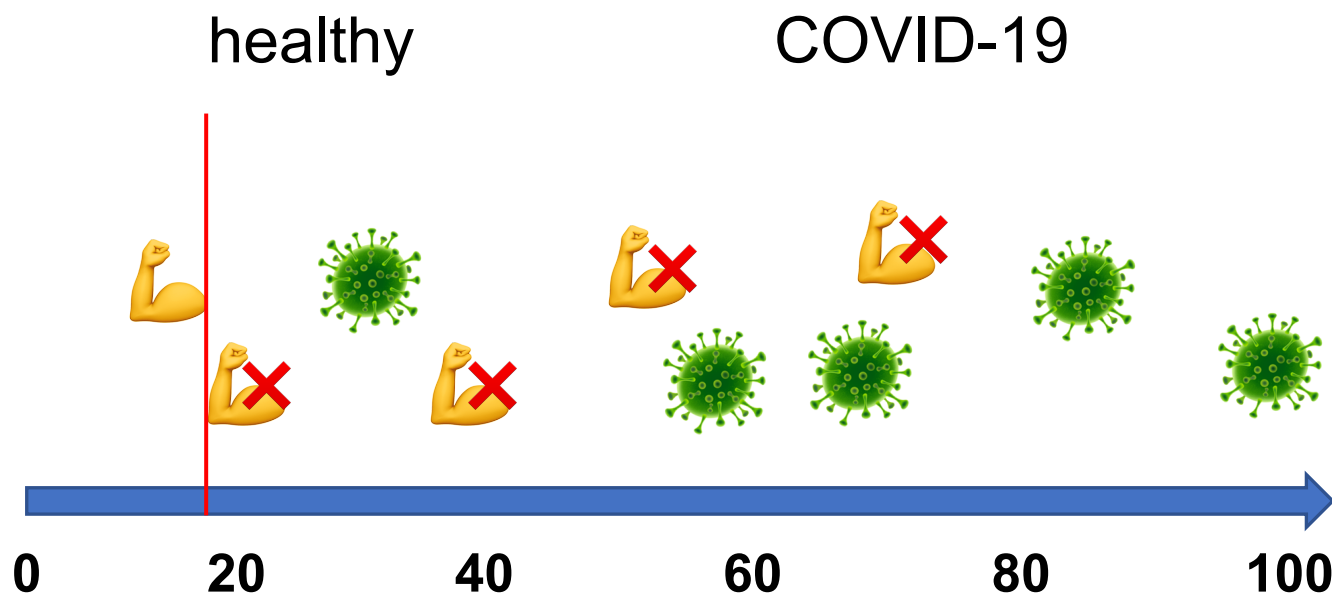
模型預測為疾病的
機率值 (%)



ROC Curve

假設我們用模型對10個人做COVID-19檢測：
(healthy 代表 negative; COVID-19 代表 positive)

TPR: 1.0, FPR: 0.8



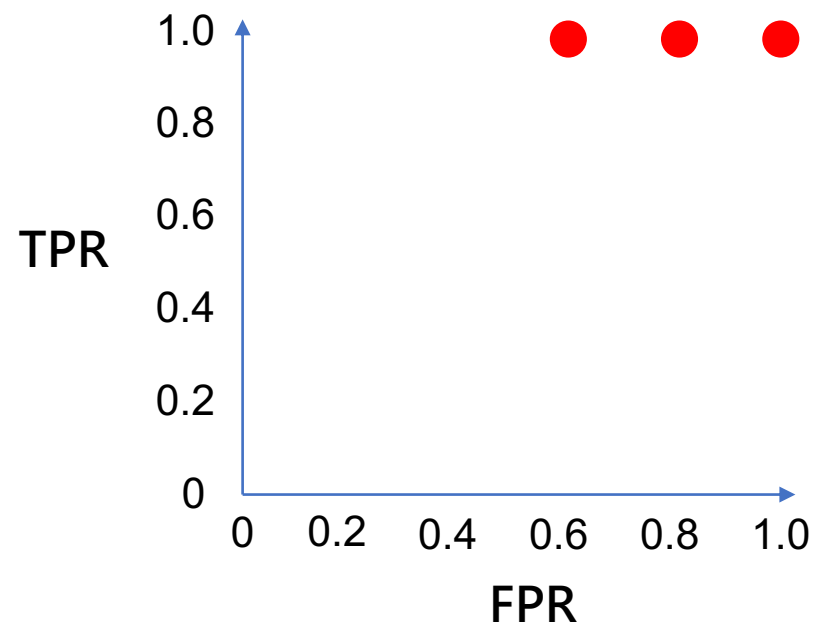
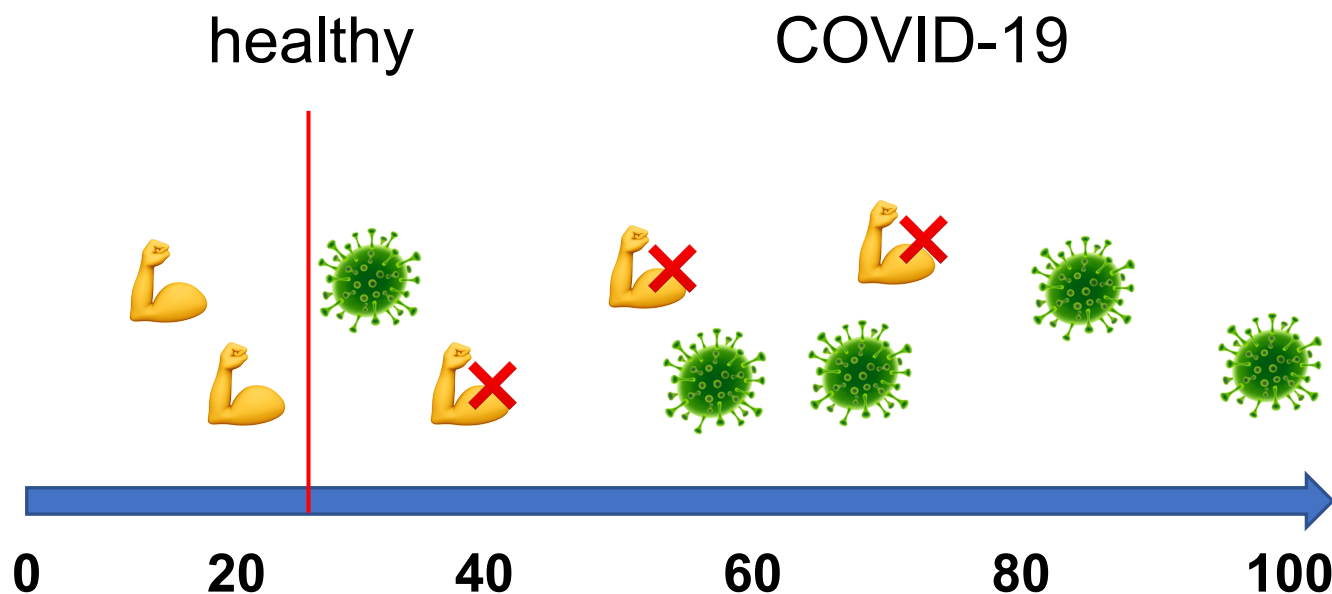
模型預測為疾病的
機率值 (%)



ROC Curve

假設我們用模型對10個人做COVID-19檢測：
(healthy 代表 negative; COVID-19 代表 positive)

TPR: 1.0, FPR: 0.6



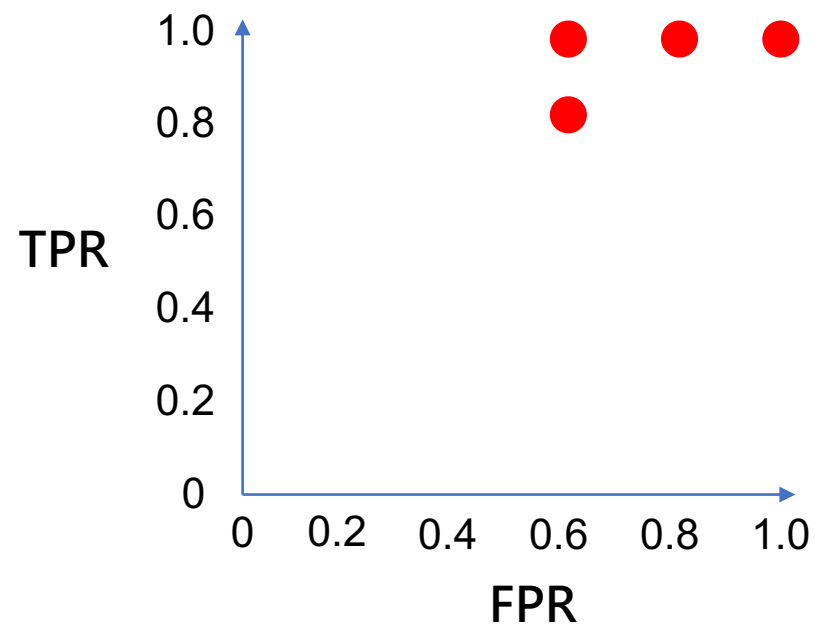
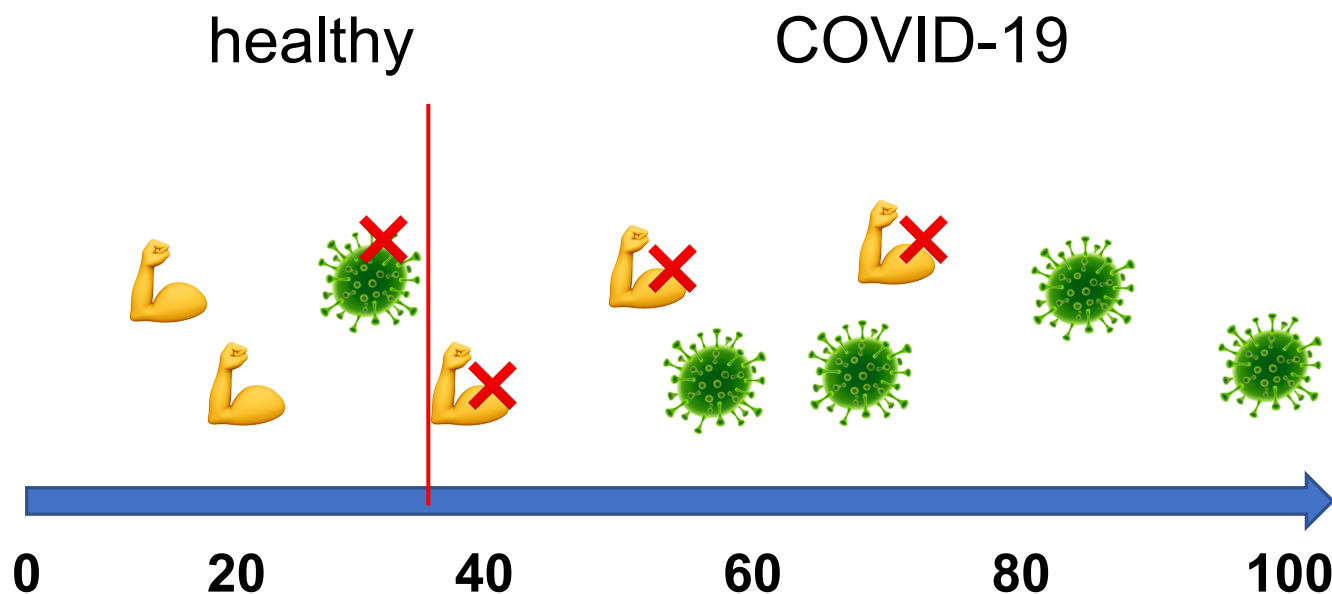
模型預測為疾病的
機率值 (%)



ROC Curve

假設我們用模型對10個人做COVID-19檢測：
(healthy 代表 negative; COVID-19 代表 positive)

TPR: 0.8, FPR: 0.6



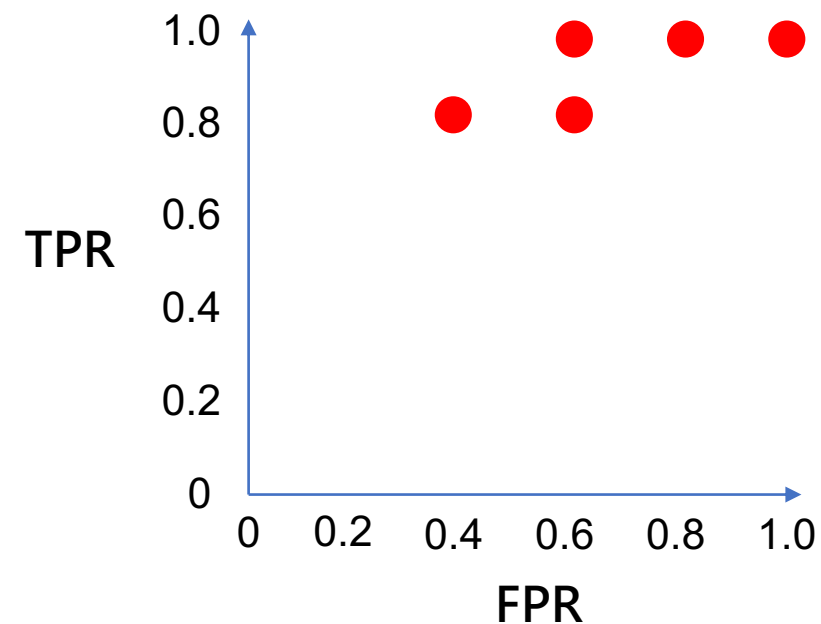
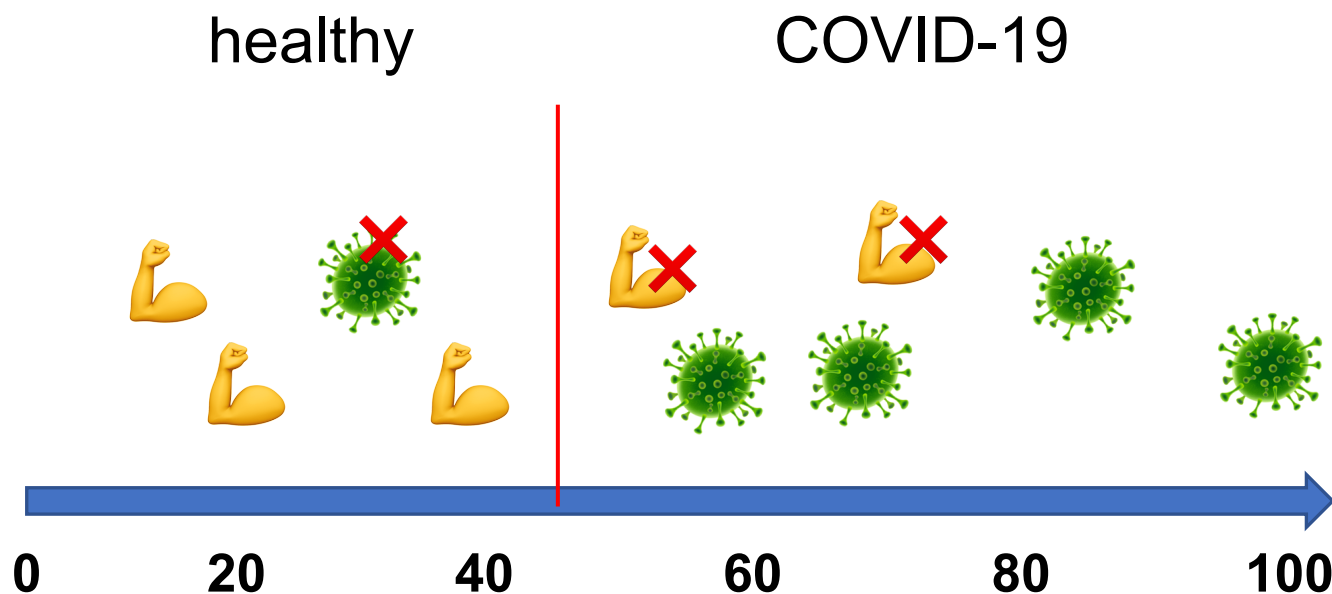
模型預測為疾病的
機率值 (%)



ROC Curve

假設我們用模型對10個人做COVID-19檢測：
(healthy 代表 negative; COVID-19 代表 positive)

TPR: 0.8, FPR: 0.4



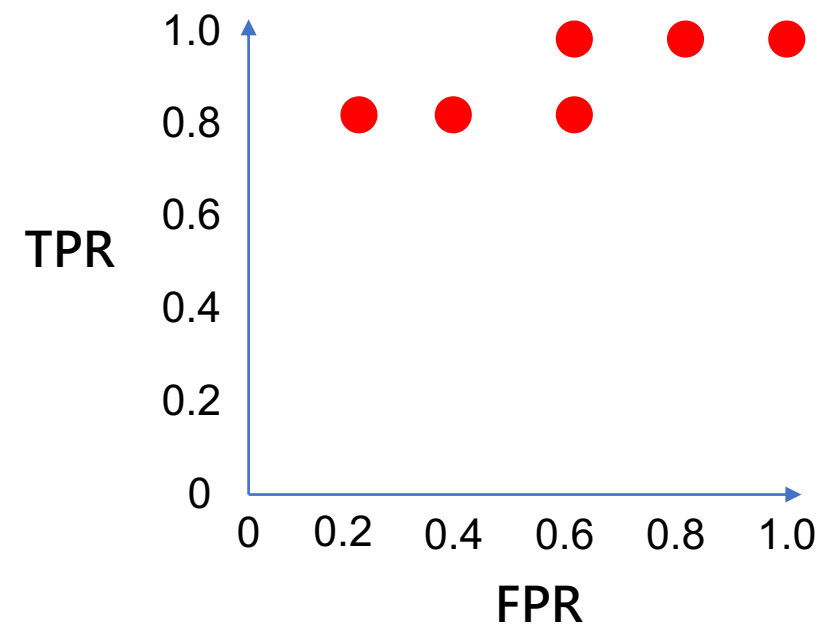
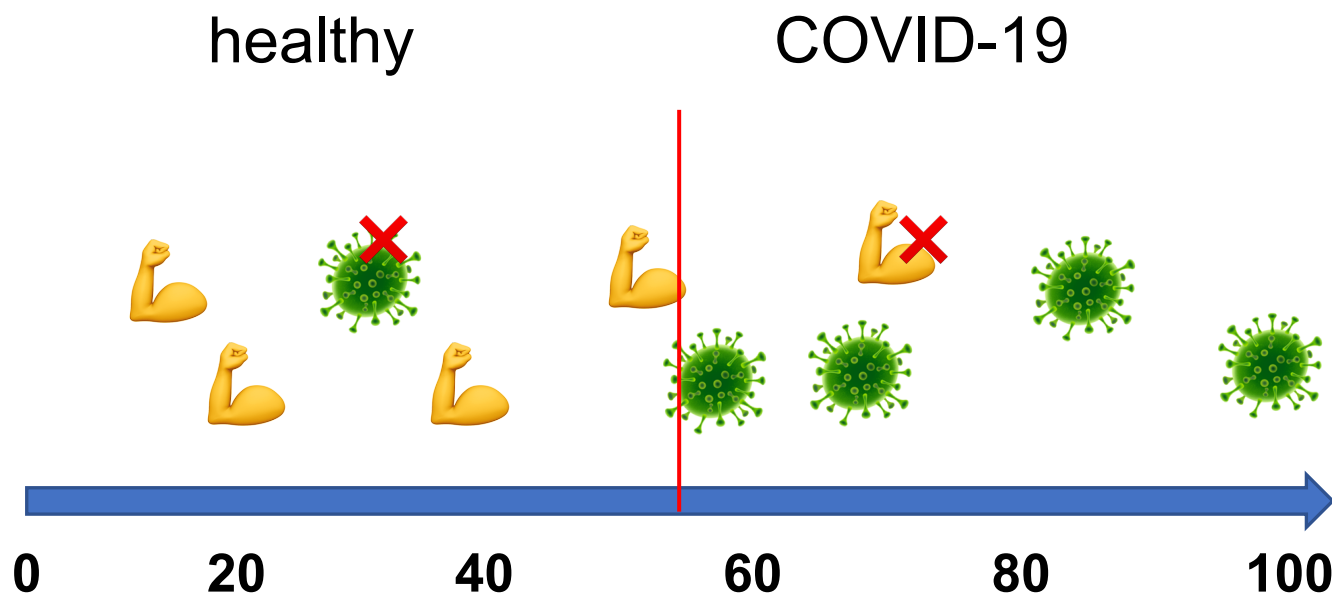
模型預測為疾病的
機率值 (%)



ROC Curve

假設我們用模型對10個人做COVID-19檢測：
(healthy 代表 negative; COVID-19 代表 positive)

TPR: 0.8, FPR: 0.2



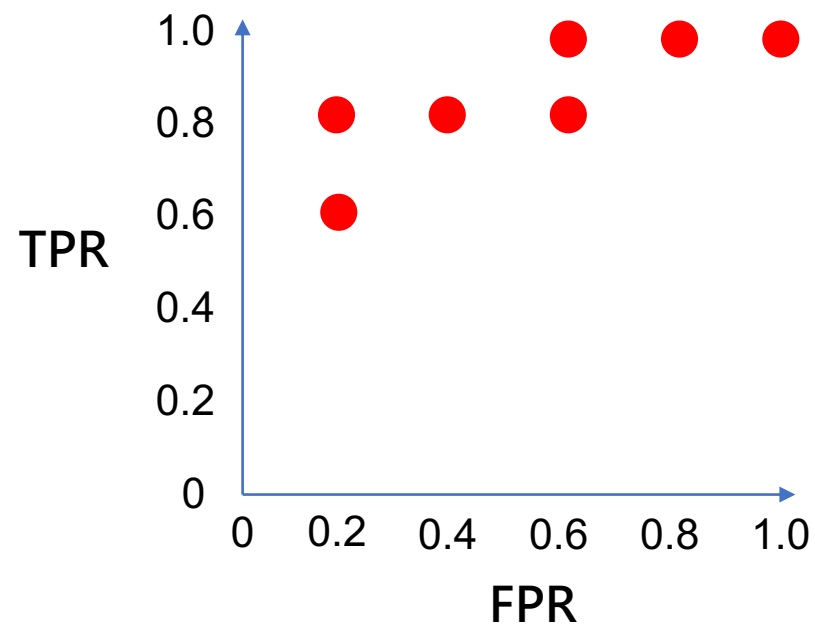
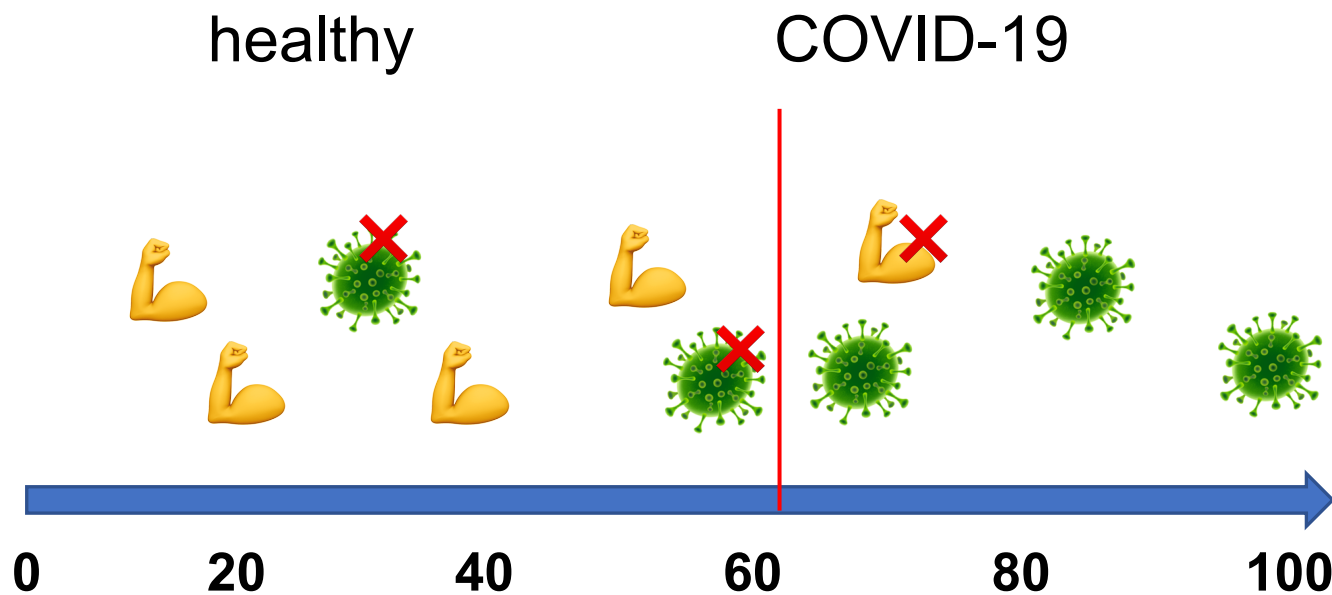
模型預測為疾病的
機率值 (%)



ROC Curve

假設我們用模型對10個人做COVID-19檢測：
(healthy 代表 negative; COVID-19 代表 positive)

TPR: 0.6, FPR: 0.2



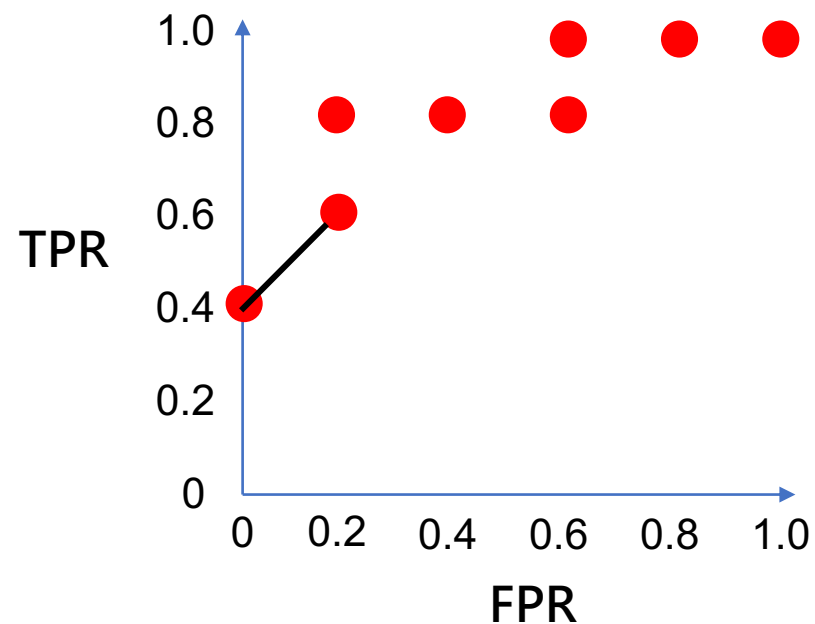
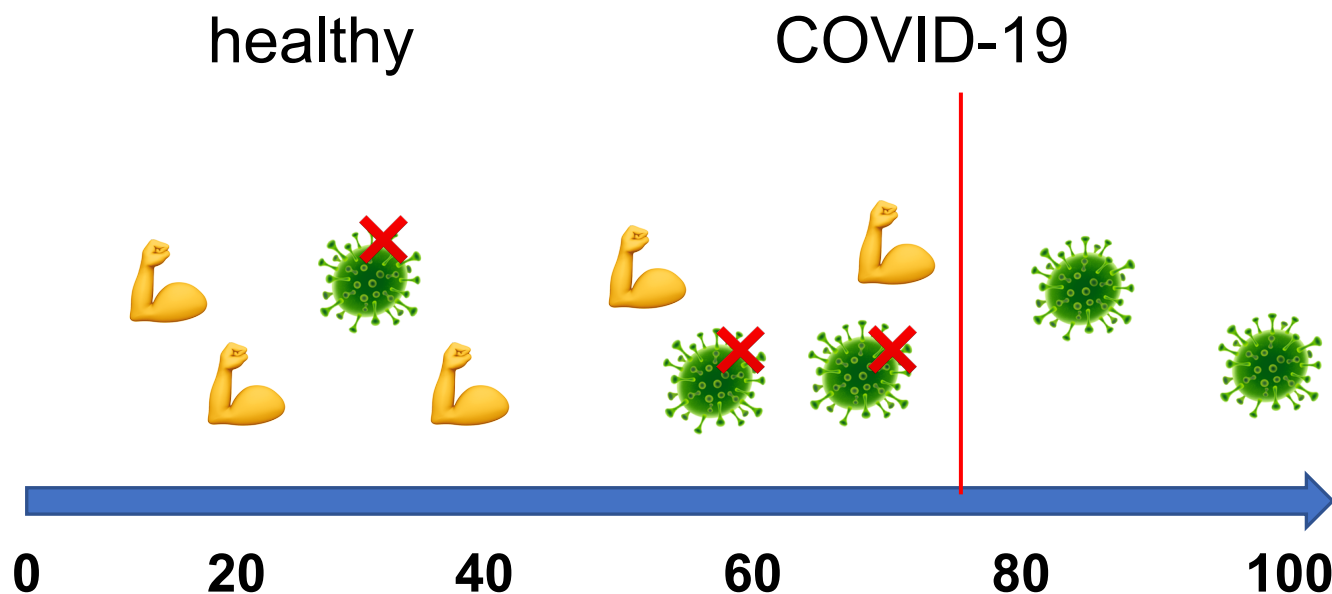
模型預測為疾病的
機率值 (%)



ROC Curve

假設我們用模型對10個人做COVID-19檢測：
(healthy 代表 negative; COVID-19 代表 positive)

TPR: 0.4, FPR: 0.0



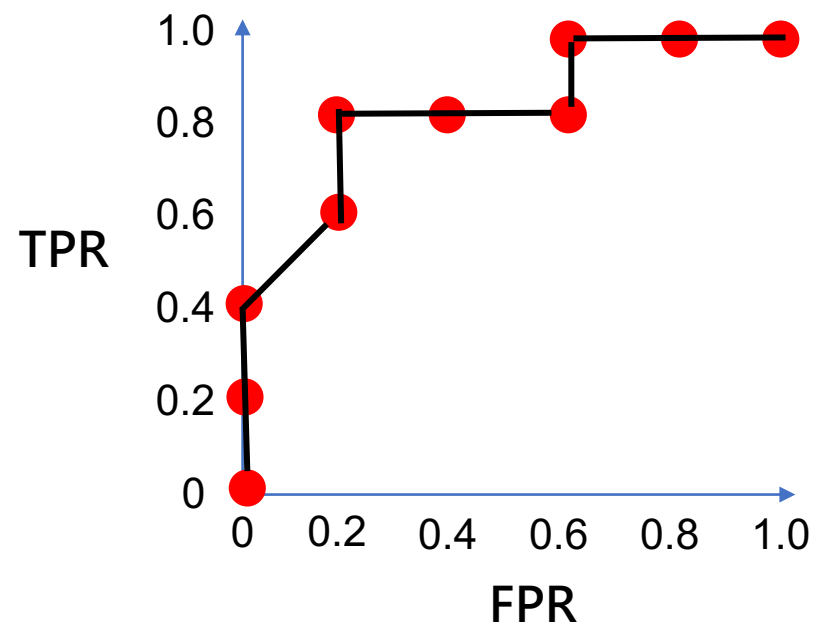
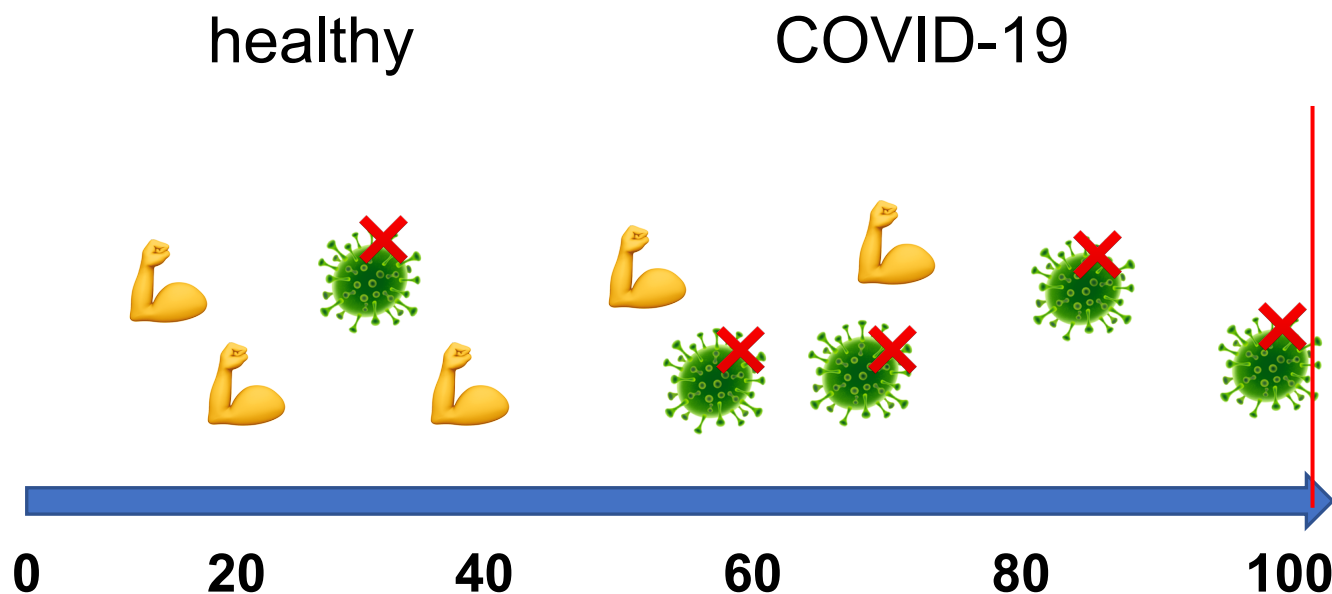
模型預測為疾病的
機率值 (%)



ROC Curve

假設我們用模型對10個人做COVID-19檢測：
(healthy 代表 negative; COVID-19 代表 positive)

TPR: 0.0, FPR: 0.0



模型預測為疾病的
機率值 (%)



ROC Curve (bad case)

(Useful!!) <https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc?hl=zh-tw>

Bad case: 模型對於 positive 和 negative 區分能力差

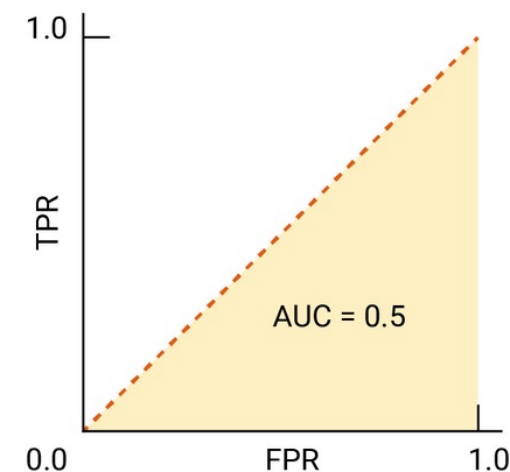
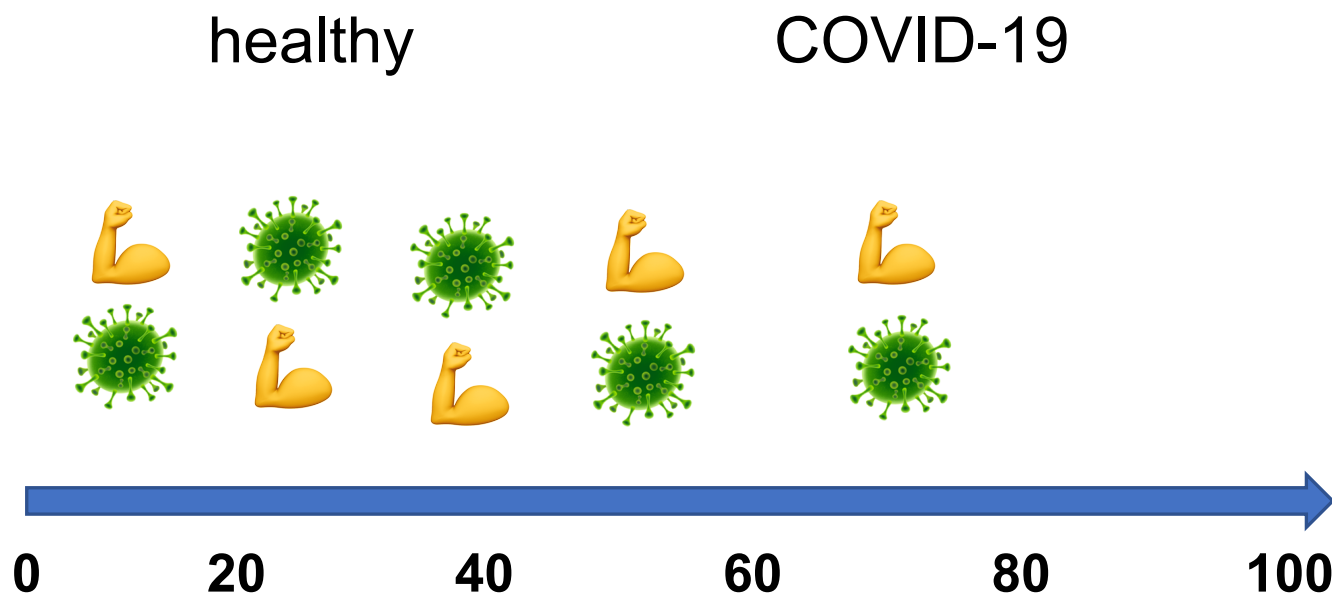


圖 2 ROC 和 AUC 完全隨機猜測。



完美模型的條件

- 分類結果完全正確
 - 沒有誤判
 - No False Negatives: TPR 越高越好
 - No False Positives: FPR 越低越好
- 完美模型中存在一個「理想 threshold 區間」，使我們能同時得到 $TPR = 1$ 且 $FPR = 0$

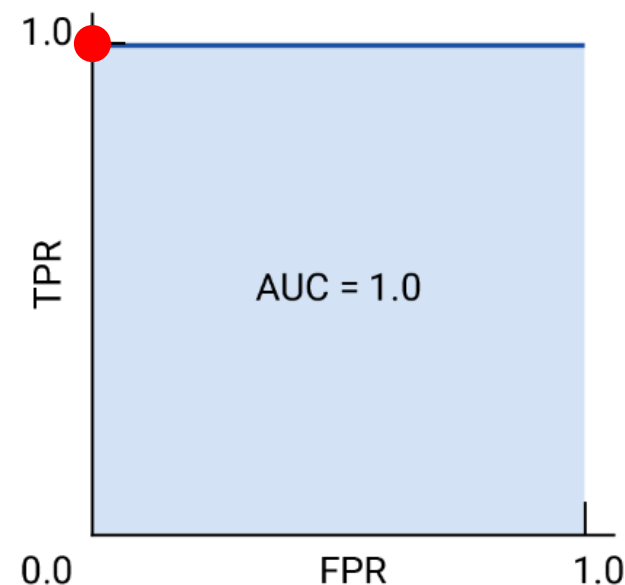


圖1：ROC 和 AUC，這是一個完美假設的模型。



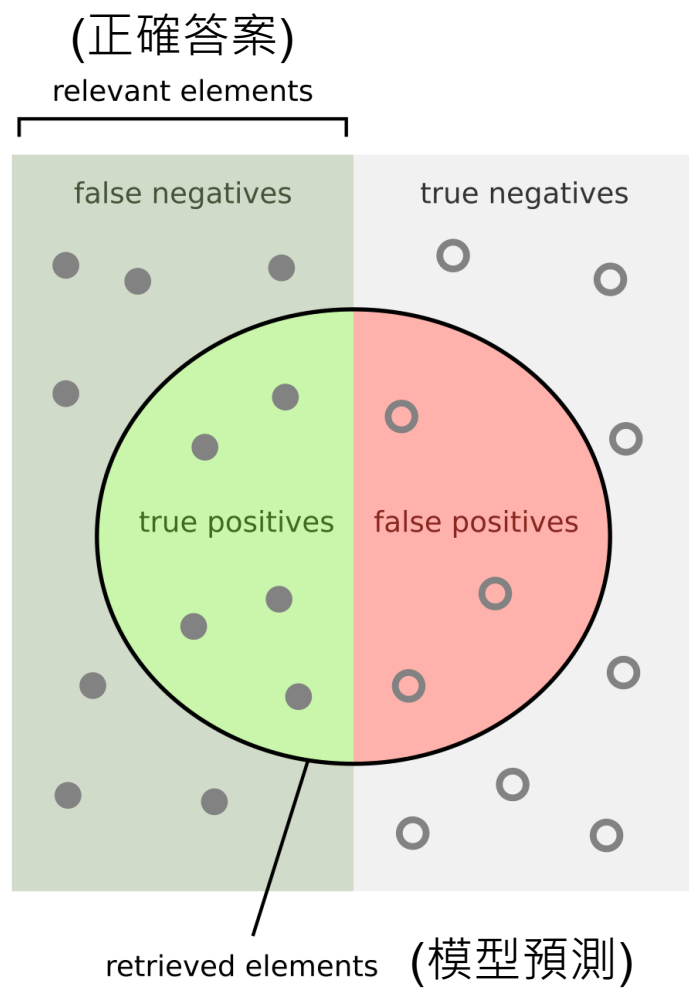
Confusion Matrix (混淆矩陣) for ROC

	Actually positive	Actually negative
Predicted positive	True positive (TP)	False positive (FP)
Predicted negative	False negative (FN)	True negative (TN)

- $TP / (TP + FN)$: True Positive Rate (TPR)
 - 又稱作 Recall
- $FP / (FP + TN)$: False Positive Rate (FPR)



Recall and Precision



How many retrieved items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

How many relevant items are retrieved?

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$



Confusion Matrix for General Uses

	Actually positive	Actually negative
Predicted positive	True positive ↑ (TP)	False positive (FP)
Predicted negative	False negative (FN)	↑ True negative (TN)

- 斜對角數值越大越好
 - TP 和 TN 越大越好
- $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$
 - 模型預測的TP比例
- $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$
- $\text{F1-score (廣義)} = 2(\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$



roc_curve using scikit-learn

- https://scikit-learn.org/stable/modules/generated/sklearn.metrics.roc_curve.html

```
import numpy as np
from sklearn import metrics

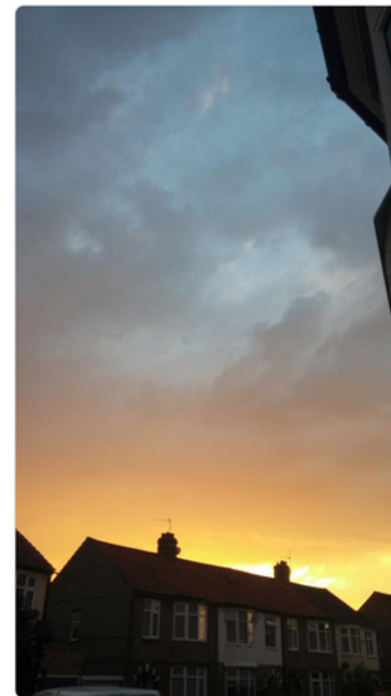
y = np.array([1, 1, 2, 2])
scores = np.array([0.1, 0.4, 0.35, 0.8])
fpr, tpr, thresholds = metrics.roc_curve(y, scores, pos_label=2)
```



Natural Language Processing with Disaster Tweets

- You are predicting whether a given tweet is about a real disaster or not. If so, predict a 1.
 - If not, predict a 0.
- 資料範例 ➡ (但任務中只需要進行文字分類)

Anna K
@AnyOtherAnnaK
On plus side LOOK AT THE SKY LAST NIGHT IT WAS ABLAZE



12:43 AM - Aug 6, 2015 - Twitter for Android

The author explicitly uses the word “ABLAZE” but means it metaphorically. This is clear to a human right away, especially with the visual aid. But it’s less clear to a machine.

In this competition, you’re challenged to build a machine learning model that predicts which Tweets are about real disasters and which one’s aren’t. You’ll have access to a dataset of 10,000 tweets that were hand classified. If this is your first time working on an NLP problem, we’ve created a [quick tutorial](#) to get you up and running.

Disclaimer: The dataset for this competition contains text that may be considered profane, vulgar, or offensive.

Acknowledgments

This dataset was created by the company figure-eight and originally shared on their

[‘Data For Everyone’ website here.](#)

Tweet source: <https://twitter.com/AnyOtherAnnaK/status/629195955506708480>



Natural Language Processing with Disaster Tweets

- Data structure and example

train.csv (987.71 kB) 📄 🗑️ ➤ 1.43 MB

Detail Compact Column 5 of 5 columns ▾

id	keyword	location	text	target
 1 10.9k	222 unique values	[null] 33% USA 1% Other (4976) 65%	7503 unique values	 0 1
48	ablaze	Birmingham	@bbcmtd Wholesale Markets ablaze http://t.co/1HYXE0HY6C	1
49	ablaze	Est. September 2012 - Bristol	We always try to bring the heavy. #metal #RT http://t.co/YAo1e0xngw	0
50	ablaze	AFRICA	#AFRICANBAZE: Breaking news:Nigeria flag set ablaze in Aba. http://t.co/2nndBGwyEi	1

sample_submission.csv
test.csv
train.csv

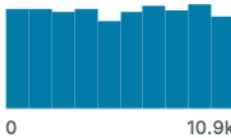


Natural Language Processing with Disaster Tweets (Submission)

- Submission 分數計算採用 F1-score

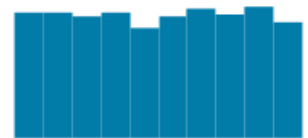
test.csv (420.78 kB)

Detail Compact Column

id	keyword	location	text
	222 unique values	[null] 34% New York 1% Other (2120) 65%	3243 unique values
0			Just happened a terrible car crash
2			Heard about #earthquake is different cities, stay safe everyone.
3			there is a forest fire at spot pond, geese are fleeing across the street, I cannot save them all
9			Apocalypse lighting. #Spokane #wildfires
11			Typhoon Soudelor kills 28 in China and Taiwan

sample_submission.csv (22.75 kB)

Detail Compact Column

id	# target
	
0	0
2	0
3	0
9	0
11	0



Wikipedia - Image/Caption Matching

<https://www.kaggle.com/competitions/wikipedia-image-caption>



→ Erantido vintra [SEP] Floro de erantido vintra

- 評估指標：NDCG@5
 - 衡量前五個排序結果的「相關性」與「排序正確性」；不只要找對，還要排得對。



Project checkpoints (暫定)

- Week 9: 確定各組的題目
- Week 11: 進度報告 PPT (5 pages)
- Week 13: 進度報告 PPT (5+5 pages), Presentations (selected teams)
- Week 15 – Week 16: Final presentations for all teams (maybe poster)
- Week 16 結束前: 繳交書面報告以及程式碼



期末 Project 規定 (暫定)

- 需要上傳 Kaggle Leaderboard
 - 這些任務都僅需要上傳 predictions 到 Kaggle 即可
- 每次報告都需要列出每位組員的貢獻內容，以及組員間的工作比重 (%)



Thank you!

Instructor: 林英嘉

 yjlin@cgu.edu.tw

TA: 劉美辰

 m1461014@cgu.edu.tw